s = (X − μ) / σ

PCA → residual

longs ≈ shorts

exit @ s→0

VIX filter

OU speed

meta-label

Δ = λ·Q

PCA-RESIDUAL STAT-ARB

Market-Neutral Statistical Arbitrage on Indian Equities

Phase 2 · Strategy Selection, Detailed Design & Block Diagram

K

Prepared by

Kartikya · Team A (Strategy Design)

THE GOAL

We want a market-neutral strategy that profits from temporary mispricings between related stocks not from guessing market direction. The bet: when a stock moves out of line with the common forces driving its peers, that deviation tends to revert.

CORE IDEA

Use PCA to strip out the common market/sector forces from every stock. What remains the residual is each stock's own idiosyncratic move. When a residual stretches too far from its equilibrium, we bet it snaps back. Short the stretched-rich names, long the stretched-cheap names.

FRAMED AS A HYPOTHESIS TEST

There is no mathematical proof that residuals revert it is an empirical hypothesis, and the edge is known to be cost-sensitive and decayed in mature markets. Per the project brief, the value is in rigorously testing whether it survives realistic Indian costs. A clean negative result is a valid result.

WHY THIS TRACK

The brief requires defending a narrow strategy choice. Here is the case.

1 Professional paradigm

Factor-removal → trade residuals is the core mid-frequency quant approach (Two Sigma / D.E. Shaw style). Based on Avellaneda-Lee (2010), a real, cited, practitioner paper.

2 Fits the time horizon

Reversions play out over hours to ~2 weeks exactly the intraday-to-short-swing window targeted, and the natural home of stat-arb.

3 Uses my strengths

PCA and portfolio construction map directly onto an IEOR / optimization background.

4 Free, accessible data

Daily adjusted OHLCV (yfinance / NSEpy) for the structure. No exotic data needed for the baseline.

5 Rich failure modes

factor structure breaks in regime shifts; costs threaten the edge exactly the kind of conditional, regime-dependent question the brief asks us to investigate.

6 Clean ML extension path

A working classical baseline first, then ML for sizing & exits follows the prescribed build order precisely.

STEP 1 — DECOMPOSE

PCA: SPLIT EACH STOCK INTO COMMON + RESIDUAL

stock return = **common part** *(market + sectors, from PCA)* + **residual** *(stock-specific leftover)*

1	**Take returns**	*60 days of daily returns for ~250–300 liquid stocks → one big matrix.*
2	**Run PCA**	*Extract the top ~5 principal components = the common forces (market, sector blocks).*
3	**Get residuals**	*For each stock: residual = its return minus the part explained by those common forces.*
4	**Model as mean-reverting**	*The cumulative residual is modelled as an Ornstein-Uhlenbeck process it wanders around an equilibrium and is pulled back, so it oscillates rather than exploding.*

STEP 2 — THE SIGNAL

THE s-SCORE: HOW STRETCHED IS THIS STOCK, RIGHT NOW?

$$s\text{-score} = \frac{\text{current residual} - \text{equilibrium}}{\text{residual's own volatility}}$$

Measured in the stock's OWN standard deviations so a calm stock and a wild stock are judged on the same scale. Not a fixed % band.

THE TRADING BANDS

$s > +2.0$	*Too RICH → SHORT it (bet it falls back)*
$-2.0 \leq s \leq +2.0$	*Normal range → no trade*
$s < -2.0$	*Too CHEAP → LONG it (bet it rises back)*

Entry threshold raised to ±2.0 (vs the literature's ±1.25) — a deliberate cost-driven choice: fewer, higher-conviction trades cut the turnover that costs punish.

AVOIDING LOOK-AHEAD

A signal computed from a closing price cannot be traded at that same close — the market is already shut. Trading on the same bar that generated the signal is look-ahead bias, the classic way backtests lie.

Bar T closes

Compute s-score from data up to and including bar T

Decision

Identify which stocks breach the band; apply filters & ML sizing

Bar T+1 opens

EXECUTE here — never on the bar that made the signal

THE RULE

Signal always leads the trade by one bar. In the daily version: signal from today's close → trade at tomorrow's open. In the intraday version: signal from the current minute → trade now. Execution modelled at the open with slippage, never at the mid.

UNIVERSE & DATA

WHICH STOCKS

- *Start from Nifty 500 a broad opportunity set.*
- *Filter hard for liquidity (volume, spread); drop penny stocks, < 2–3 yrs history, pending M&A.*
- *Left with ~250–300 clean, liquid names → 20–40 positions on a typical day.*
- *PCA uses ALL of them to find common forces; the s-score band selects which to actually trade.*

Note: even the pure-intraday version needs the daily data for its structure — two frequencies, two jobs.

WHAT DATA

STRUCTURE (slow)

60 days of daily ADJUSTED OHLCV → PCA forces + residual normal range. Free (yfinance / NSEpy).

LIVE SIGNAL (fast)

1–5 min OHLCV bars for the current residual. Paid / limited (Kite Connect; yfinance free tier to prototype).

BACKTEST WINDOW

5+ years spanning bull, COVID crash & recovery — so the model isn't biased to one regime.

ENTRY & EXIT

ENTRY — ALL GATES MUST PASS

✓ **s-score past ±2.0**	*the dislocation is extreme enough to bet on*
✓ **OU reversion filter**	*only trade stocks whose residual actually reverts (fast enough); extreme + no reversion = probably real news, skip*
✓ **Liquidity check**	*tradeable now: not halted, normal volume, reasonable spread*
✓ **Market-neutral build**	*balance longs ≈ shorts; prioritise the most extreme signals up to the position cap*

EXIT — TRIPLE BARRIER (whichever hits first)

PROFIT

s-score reverts to ±0.5 → take profit

STOP

s-score diverges to ±3.0 → cut the loss (critical: thesis was wrong)

TIME

held ~15 days without reverting → free the capital

SIZING · REGIME · RISK

final size = base risk (vol-scaled) × ML multiplier × regime multiplier *(bounded by hard caps)*

SIZING

- *Equal RISK, not equal rupees size inversely to volatility.*
- *Calm stock → bigger; wild stock → smaller. Each position = equal risk.*
- *Total portfolio targeted at ~1% daily vol.*

REGIME

- *India VIX throttle, graduated:*
- *< 18 full · 18–25 half · 25–35 quarter · > 35 stop.*
- *Scales ALL positions down as danger rises softens crashes (reacts slightly late).*

RISK

- *Layered hard limits: 5%/stock · 2× gross · net≈0 · ≤40 pos · ≤30%/sector.*
- *Drawdown breaker: halve @5% · no-new @10% · close-all @15%.*
- *Never ML-controlled safety stays non-negotiable.*

Stat-arb's hidden risk: in a stress event, "diversified" positions can become one correlated bet (the 2007 quant quake). The defenses soften this — they cannot eliminate it.

COSTS — THE CONSTRAINT

This strategy is high-turnover and cost-sensitive. Costs are not a footnote they may decide whether it profits at all. They are modelled PESSIMISTICALLY.

STT (delivery)	**~0.20%**	*the dominant India-specific cost — both sides*
STT (intraday)	**~0.025%**	*much lower — but forces same-day exit*
Brokerage	**~0.02–0.05%**	*discount-broker flat / small %*
Bid-ask spread	**~0.05–0.10%**	*paid every round trip — liquidity filter helps*
Market impact	**~0.05–0.10%**	*own order moves price (Kyle's λ)*

≈ 0.35 – 0.50% per round trip

THE HARD MATH

Gross edge per trade is small (~0.3–0.8% averaged over winners & losers). After ~0.4% costs, net edge ranges from slightly negative to modestly positive. Costs consume most of the gross edge.

Levers: raise threshold · hold longer · trade only liquid names · scalp fast reversions intraday at low STT.

If costs kill it, that is a legitimate, valuable result — exactly what the brief asks us to establish honestly.

THE ML LAYER

Built ONLY after the classical baseline works and each piece must be proven to beat it. ML refines SIZING and TIMING; it never picks direction or overrides safety.

1

META-LABELLING → SIZING

A gradient-boosting model takes situational features (s-score, OU speed, residual age, volatility, volume, PCA fit, VIX, trend) and outputs P(this trade works). That probability scales the position. Trained on triple-barrier outcomes, sample-weighted by P&L magnitude so big winners/losers count more.

2

ML EXIT MODEL

Given an open trade's state (current s-score, days held, profit-after-cost, regime), decide exit vs hold smarter than fixed barriers. Trained on what historical paths reveal as the better action. Guardrail: the hard stop loss is ALWAYS active; the model can exit early but can never hold a runaway loser.

EVALUATION

How we test honestly without fooling ourselves. This is where the project's real value lives.

WALK-FORWARD

Train on the past, test on the future, roll forward. Mimics reality; tests every regime. Gold standard for time series.

NET METRICS

Net Sharpe (after costs — the number that matters), max drawdown & duration, win rate, avg win vs loss, turnover.

DEFLATED SHARPE

Correct the Sharpe for how many variations were tried stops us reporting the luckiest of many attempts as skill.

REGIME BREAKDOWN

Score calm / crisis / choppy separately. Did it survive COVID? Did the regime filter actually protect it?

ONE HONEST LOOK

Build & tune only on dev data. One look at held-back test data no tweaking after. That estimate is the truth.

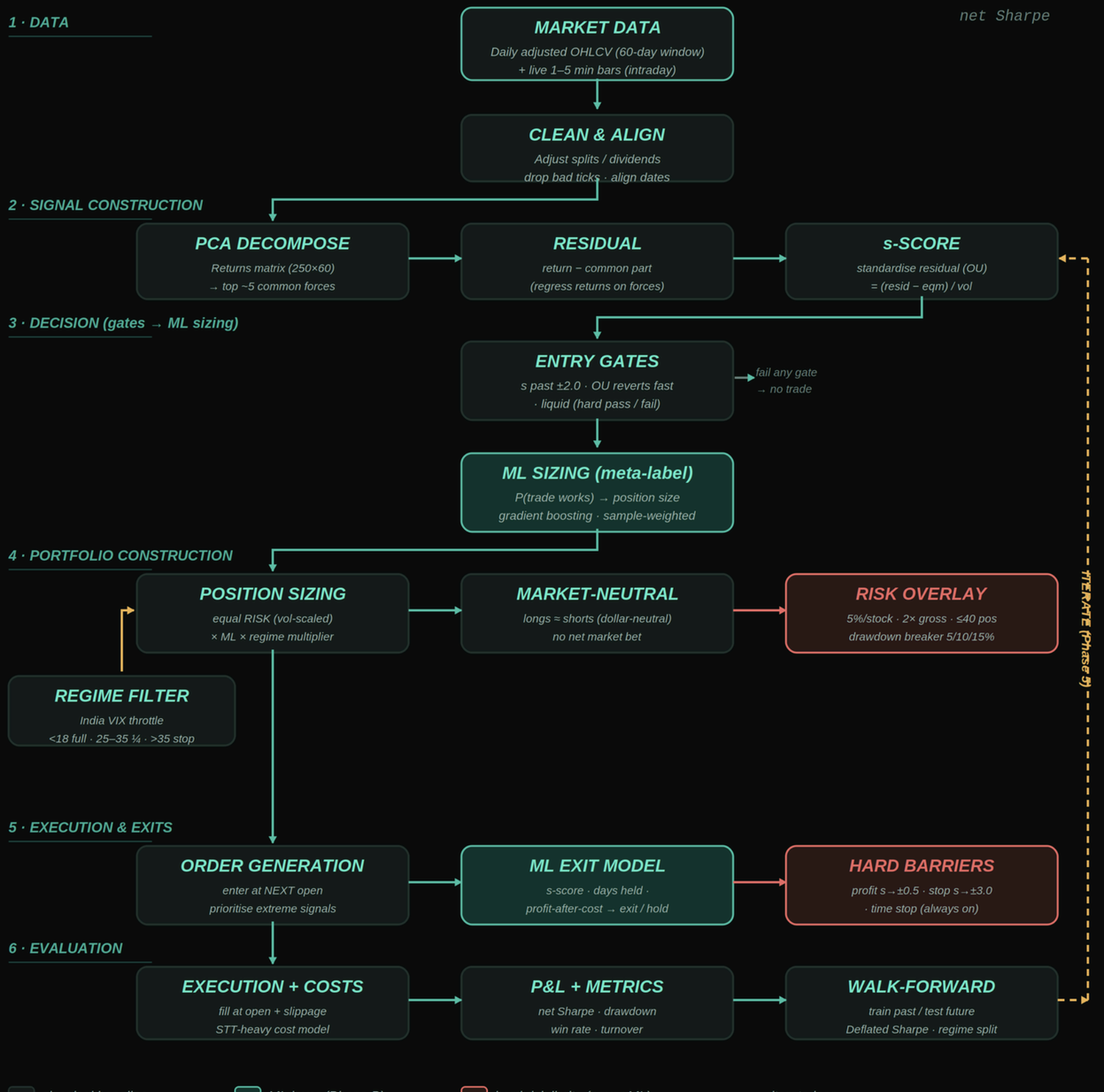
REPORT FAILURES

Net performance, drawdowns, where it broke, number of variations. Honesty over inflated metrics, per the brief.

STRATEGY BLOCK DIAGRAM

PCA-Residual Statistical Arbitrage · information flow & decision-making

```
s = (X-μ)/σ
PCA → residual
longs ≈ shorts
OU revert
net Sharpe
```



ROADMAP

BUILD ORDER (per the brief)

1	**Classical baseline**	*PCA-residual s-score, fixed rules, end-of-day. Prove it works (or fails) alone.*
2	**Add ML sizing**	*Meta-labelling for position size. Must beat the baseline.*
3	**Add ML exits**	*Context-aware exits, hard stop always on. Must beat fixed barriers.*
4	**Go intraday**	*Live minute s-score, low-STT scalping of fast reversions.*
5	**Iterate**	*Backtest → analyse → refine filters & risk. Document every step.*

NOT YET DESIGNED (optional): *signal combination (only if multiple signals exist) and a news-sentiment / agentic layer (GDELT + FinBERT). Flagged as exploratory — deliberately not yet specified.*

ONE-LINE SUMMARY

A market-neutral PCA-residual stat-arb strategy on ~250–300 liquid Indian stocks: strip out common forces, trade the mean-reversion of the leftover, hold hours-to-weeks built classical-first, then refined with ML sizing & exits and an intraday extension, and judged honestly after realistic costs.

THE HONEST POSITION

This is a real, professional strategy and a genuinely hard one to make profitable under Indian retail costs. The project's worth is not a guaranteed return; it is a rigorous, cost-honest test of whether the edge survives, with a clean negative result treated as a valid finding.

THANK YOU